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ISM
2022-23 EC2M-Jan

1) b) If 2 events are mutually exclusive, then they are independent also & vice versa.

Mutual Exclusive: $P(A \cap B) = 0$ $P(A \cup B) = P(A) + P(B)$

Mutually independent: $P(A \cap B) = P(A)P(B)$

$$P(A|B) = P(A)$$

$$P(B|A) = P(B)$$

Mutually exclusive $\Rightarrow$ mutually independent

2 events can't happen together, so their occurrence doesn't affect each other.

$\therefore$ They are indep.

~~Eg: flipping coin & rolling dice.~~

MI $\nRightarrow$ ME

eg: flipping coin & rolling dice
independent but can happen together.

2) a) $P(A) = \frac{1}{2}$ $P(B) = \frac{1}{3}$ $P(A|B) = \frac{1}{8} = \frac{P(A \cap B)}{P(B)}$

i) $P(B|A) = \frac{1/18}{1/2} = 1/9$

ii) $P(B|A') = \frac{\frac{1}{3} - \frac{1}{18}}{1 - 1/2} = 5/9$

iii) $P(A \cup B|A) = 1$ ($P(A \cup B|A) = \frac{P(A \cup B \cap A)}{P(A)} = \frac{P(A)}{P(A)} = 1$)

iv) $P(B|A) = 1/9$

v) $P(A \cap B) = \frac{1}{2} - \frac{1}{18} = 4/9$

vi) $P(\bar{A} \cap \bar{B}) = 1 - 7/9 = 2/9$

vii) $P(\bar{A} \cap B) = \frac{1}{3} - \frac{1}{18} = 5/18$

$$P(A \cap B) = P(A|B)P(B) \\ = \frac{1}{6} \times \frac{1}{3} \\ = \frac{1}{18}$$

$$P(A \cup B) = P(A) + P(B) \\ - P(A \cap B) \\ = \frac{1}{2} + \frac{1}{3} - \frac{1}{18} \\ = 7/9$$

2) b)

$P(A) = 15\%$ (self)
 $P(B) = 25\%$ (bad road)
 $P(C) = 20\%$ (good discipline)
 $P(D) = 40\%$ (other)

$P(E|A) = 1\%$
 $P(E|B) = 2\%$
 $P(E|C) = 3\%$
 $P(E|D) = 4\%$

100% ✓

E : death after accident

$$\begin{aligned}
 P(E) &= 0.15 \times 0.01 + 0.25 \times 0.02 + 0.2 \times 0.03 + 0.4 \times 0.04 \\
 &= \frac{57}{2000} = 0.029 \\
 &= \boxed{2.9\%}
 \end{aligned}$$

3) a) X = random variable

$$f(x) = \begin{cases} k e^{-0.5x} & x \geq 0 \\ 0 & \text{else} \end{cases}$$

i) value of k

$$\int_{-\infty}^{\infty} f(x) dx = \underbrace{\int_{-\infty}^0 f(x) dx}_{=0} + \int_0^{\infty} f(x) dx = 1$$

$$= \int_0^{\infty} k e^{-0.5x} dx = k \int_0^{\infty} e^{-0.5x} dx = 1$$

$$= \left[\frac{-k e^{-0.5x}}{0.5} \right]_0^{\infty} = 1$$

$$\Rightarrow k \cdot \frac{2}{1}$$

$$\Rightarrow \left[\frac{-k e^0}{0.5} \right] + \left[\frac{-k e^{-\infty}}{0.5} \right] = 1$$

$$\Rightarrow \boxed{k = 0.5}$$

$$ii) P(-1.5 < X < 2.5)$$

$$= P(-1.5 < X \leq 0) + P(0 < X < 2.5)$$

$$= \frac{1}{2} \int_0^{2.5} e^{-0.5x} dx$$

$$= \frac{1}{2} \times \left[\frac{e^{-0.5x}}{-0.5} \right]_0^{2.5}$$

$$= \frac{1}{2} \times \left[e^{-0.5 \times 2.5} - 1 \right]$$

$$= \boxed{0.713}$$

$$iii) P([X]=1) = P(1 \leq X < 2) = -[e^{-0.5x}]_1^2$$

$\downarrow$
P(floor)

$$= e^{-0.5} - e^{-1}$$

$$= \boxed{0.239}$$

3b)

$P(x,y)$		X	
		1	2
Y	1	0.15	0.15
	2	0.2	0.2
	3	0.05	0.1
	4	0.05	0.1

i) $\sum_{x,y} P(x,y) = 1$ $\therefore$ Distribution is right.

ii) Marginal distribution of X & Y :

	1	2
X	0.45	0.55

	1	2	3	4
Y	0.3	0.4	0.15	0.15

iii) $P(Y=2 | X < 2) = \frac{P(Y=2 \cap X < 2)}{P(X < 2)} = \frac{0.2}{0.45} = \frac{4}{9} = 0.444$

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$$iv) P(X=1 \text{ AND } Y=1) = 0.15$$

$$P(X=1) = 0.45$$

$$P(Y=1) = 0.3$$

$$P(X=1)P(Y=1) = 0.135 \neq P(X=1 \text{ and } Y=1)$$

$\therefore$ not independent.

4) a) binomial distribution

$$\text{mean} = np = 4$$

$$n = \frac{4}{\frac{1}{3}} = 6$$

$$\text{variance} = np(1-p) = \frac{4}{3}$$

$$1-p = \frac{1}{3} \quad p = \left(\frac{2}{3}\right)$$

$$n = 6$$

$$p = \frac{2}{3}$$

$$P(X=2) = {}^6C_2 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^{6-2}$$

$$b) \text{ mean} = \mu = 250 \text{ kg}$$

$$Z = \frac{x - 250}{\sqrt{20}}$$

$$\text{std dev} = \sigma = 20 \text{ kg}$$

~~mean = 250 kg~~

$$i) P(X > 200) = 1 - P\left(Z > \frac{200 - 250}{20}\right) = 1 - P(Z > -2.5) \\ = 1 - 0.00621 = \boxed{0.99379}$$

$$ii) P(250 < X < 300) = P\left(Z < \frac{300 - 250}{20}\right) - P\left(Z < \frac{250 - 250}{20}\right) \\ = P(Z < 2.5) - P(Z < 0) \\ = 0.9938 - 0.5 \\ = \boxed{0.4938}$$

$$5) \quad n = 35 \\ \mu = 15 \\ \text{sd} = \sigma = 2$$

assuming: random sampling & normal distribution

$$SE = \frac{\sigma}{\sqrt{n}} = 0.34$$

$$Z_1 = \frac{12 - \mu}{SE} = -8.82 \quad P(Z < Z_1) \approx 0$$

$$Z_2 = \frac{17 - \mu}{SE} = 5.88 \quad P(Z < Z_2) = 0.9987$$

$$P(Z_1 < Z < Z_2) = P(Z < Z_2) - P(Z < Z_1) = \boxed{0.9987}$$